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Studierichting: Informatica
Oefeningenreeks 4A nr. 42.9

$$\int \frac{5x-2}{x^2-4}\,dx$$

$$x^2-4=(x-2)(x+2)$$

$$\frac{5x-2}{x^2-4}=\frac{5x-2}{(x-2)(x+2)}=\frac{A}{x-2}+\frac{B}{x+2}$$

$$A=\left.\frac{5x-2}{x+2}\right|_{x=2}=\frac{8}{4}=2$$

$$B=\left.\frac{5x-2}{x-2}\right|_{x=-2}=\frac{-12}{-4}=3$$

$$\frac{5x-2}{x^2-4}=\frac{2}{x-2}+\frac{3}{x+2}$$

$$\int \frac{5x-2}{x^2-4}\,dx=2\int\frac{dx}{x-2}+3\int\frac{dx}{x+2}$$

$$=2\ln|x-2|+3\ln|x+2|+c$$

References